

# **B.TECH (CIVIL ENGINEERING) – 8th SEMESTER**

## **ENVIRONMENTAL ENGINEERING – ASSIGNMENT**

### **Topic: Design of a Simple Water Treatment Plant in Seville**

**Submitted by:** Dilshada Farooq (210324)

#### ***Objective:***

To study and design a simple water treatment plant suitable for the city of Seville, ensuring safe and sustainable water supply for domestic and municipal use.

#### ***Introduction:***

Water treatment is an essential process in civil and environmental engineering. The goal is to provide clean, potable water free from harmful microorganisms and impurities. In Seville, a Mediterranean city characterized by warm temperatures and moderate rainfall, water treatment systems must be designed to handle seasonal variations and ensure efficiency in purification processes.

#### ***Design Process:***

A simple water treatment plant generally consists of the following major units: 1. **Intake Structure:** Collects raw water from a river, lake, or reservoir. 2. **Screening:** Removes large floating debris like leaves, branches, and plastics. 3. **Coagulation and Flocculation:** Addition of coagulants (like alum) to aggregate suspended particles into larger flocs. 4. **Sedimentation:** Allows heavy flocs to settle at the bottom, clarifying the water. 5. **Filtration:** Water passes through sand or gravel filters to remove finer particles. 6. **Disinfection:** Usually done with chlorine or UV treatment to kill remaining microorganisms. 7. **Storage and Distribution:** Treated water is stored in reservoirs and distributed through pipelines. These steps ensure that water meets WHO and local water quality standards.

#### ***Design Parameters and Calculations:***

For Seville, the design parameters depend on population size, water demand, and source quality. Assuming an average per capita consumption of 150 liters per day and a population of 100,000: - **Total daily demand:** 15,000,000 liters/day (15 MLD) - **Design flow rate:** 0.174 m<sup>3</sup>/s The plant units are designed as follows: - **Sedimentation tank:** 3–4 hours detention time, surface loading rate of 30 m<sup>3</sup>/m<sup>2</sup>/day. - **Filter units:** Dual media filters with 0.6 m sand depth and 0.45 m anthracite layer. - **Chlorination:** 1–2 mg/L of chlorine dose for effective disinfection.

#### ***Advantages of the Design:***

1. Ensures safe and potable water supply. 2. Simple design with low maintenance requirements. 3. Adaptable to local climatic and water quality conditions. 4. Can be scaled based on community size and demand.

***Conclusion:***

The design of a simple water treatment plant for Seville demonstrates the integration of basic civil engineering principles with environmental sustainability. The plant design ensures a consistent supply of safe drinking water by employing effective treatment processes. With appropriate maintenance and monitoring, such a system can serve as a reliable and economical solution for urban and semi-urban communities.

***Result:***

The proposed water treatment plant design effectively treats raw water to meet drinking water standards and ensures continuous supply to the residents of Seville.